

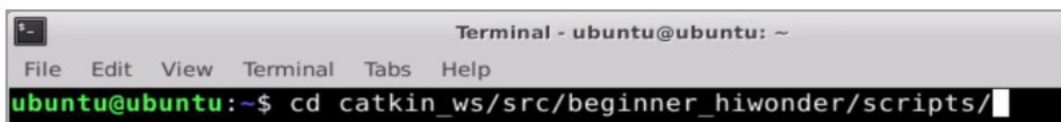
第6课 编写简单的订阅器 Subscriber

订阅器是基于编辑了发布器的基础上创建的，只有发布了消息，才有可能订阅。若未编辑发布器，可前往目录“第2章 ROS开发入门->ROS基础课程->第4课 编辑简单的发布器 Publisher”查看编辑教程。

1. 编写订阅者节点

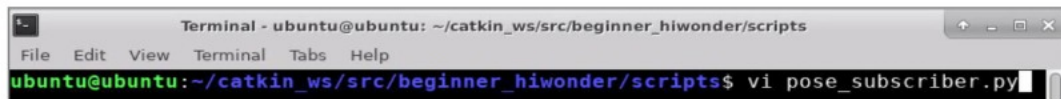
这里我们以创建一个的pose_subscriber.py节点为例进行讲解。

- 1) 输入指令“`cd catkin_ws/src/beginner_hiwonder/scripts/`”，回车。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ cd catkin_ws/src/beginner_hiwonder/scripts/
```

- 2) 输入指令“`vi pose_subscriber.py`”编辑程序，复制下面程序。如需修改，再按下“i”即可修改。修改完成，按下“Esc”，输入“: wq”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ vi pose_subscriber.py
```

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

import rospy

from turtlesim.msg import Pose

def poseCallback(msg):

    rospy.loginfo("Turtle pose: x:%0.6f, y:%0.6f", msg.x, msg.y)
```

```
def pose_subscriber():

    # ROS节点初始化

    rospy.init_node('pose_subscriber', anonymous=True)

    # 创建一个Subscriber，订阅名为/turtle1/pose的topic，注册回调函数poseCallback

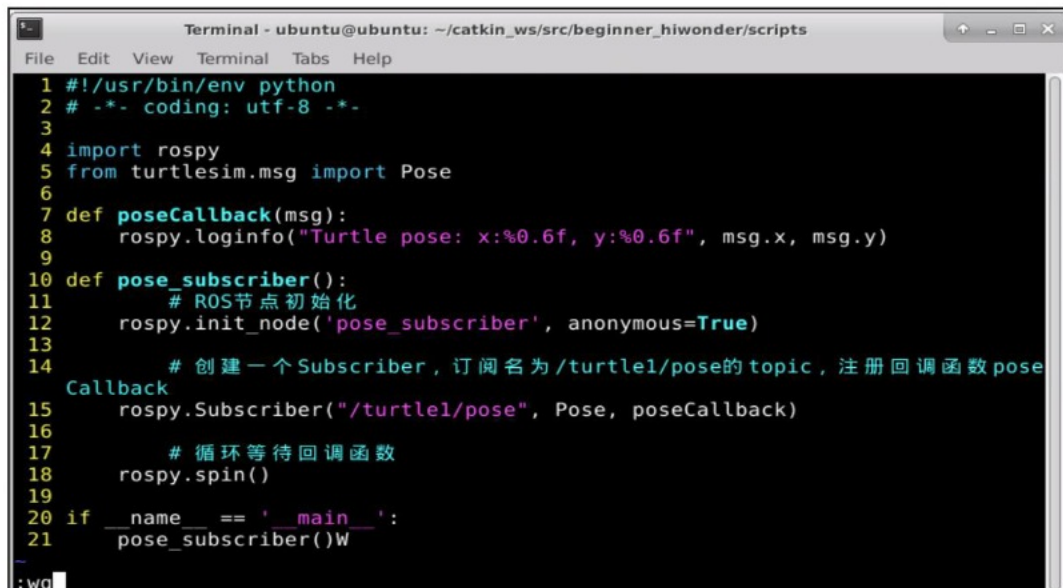
    rospy.Subscriber("/turtle1/pose", Pose, poseCallback)

    # 循环等待回调函数

    rospy.spin()

if __name__ == '__main__':

    pose_subscriber()
```

A terminal window titled "Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts" displays the execution of a Python script. The script content is as follows:

```
1 #!/usr/bin/env python
2 # -*- coding: utf-8 -*-
3
4 import rospy
5 from turtlesim.msg import Pose
6
7 def poseCallback(msg):
8     rospy.loginfo("Turtle pose: x:%0.6f, y:%0.6f", msg.x, msg.y)
9
10 def pose_subscriber():
11     # ROS节点初始化
12     rospy.init_node('pose_subscriber', anonymous=True)
13
14     # 创建一个Subscriber，订阅名为/turtle1/pose的topic，注册回调函数poseCallback
15     rospy.Subscriber("/turtle1/pose", Pose, poseCallback)
16
17     # 循环等待回调函数
18     rospy.spin()
19
20 if __name__ == '__main__':
21     pose_subscriber()W
```

The terminal shows the script being executed line by line, with the prompt character changing from '#' to '\$' at the end of each line. The prompt at the bottom is ':wq'.

3) 输入指令“**chmod +x pose_subscriber.py**”回车，为保存的pose_subscriber.py赋予可执行权限。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/beginner_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x pose_subscriber.py
```

2. 测试发布者和订阅者

1) 输入指令“**roscore**”，启动节点管理器。



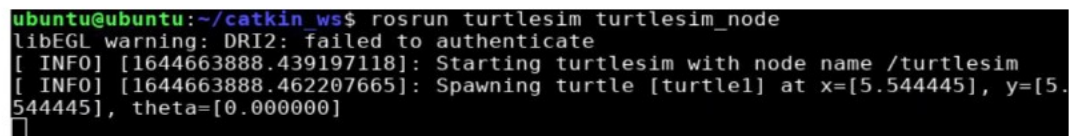
```
ubuntu@ubuntu:~/catkin_ws$ roscore
```

若已开启，则会出现以下提示：

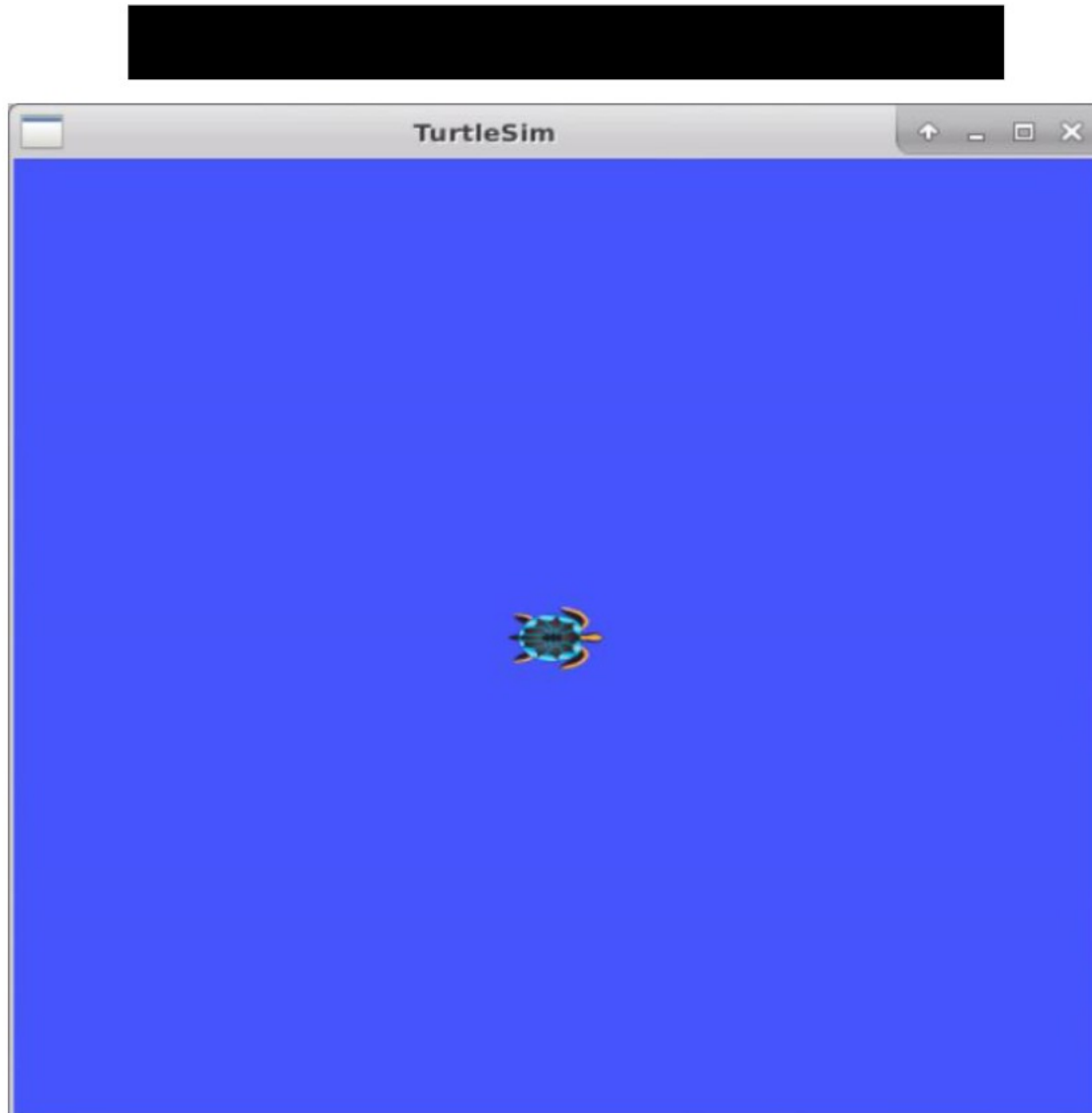


```
RLException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://ubuntu:11311/
The traceback for the exception was written to the log file
```

2) 再输入指令“**roslaunch turtlesim turtlesim_node**”，回车，开启小乌龟。

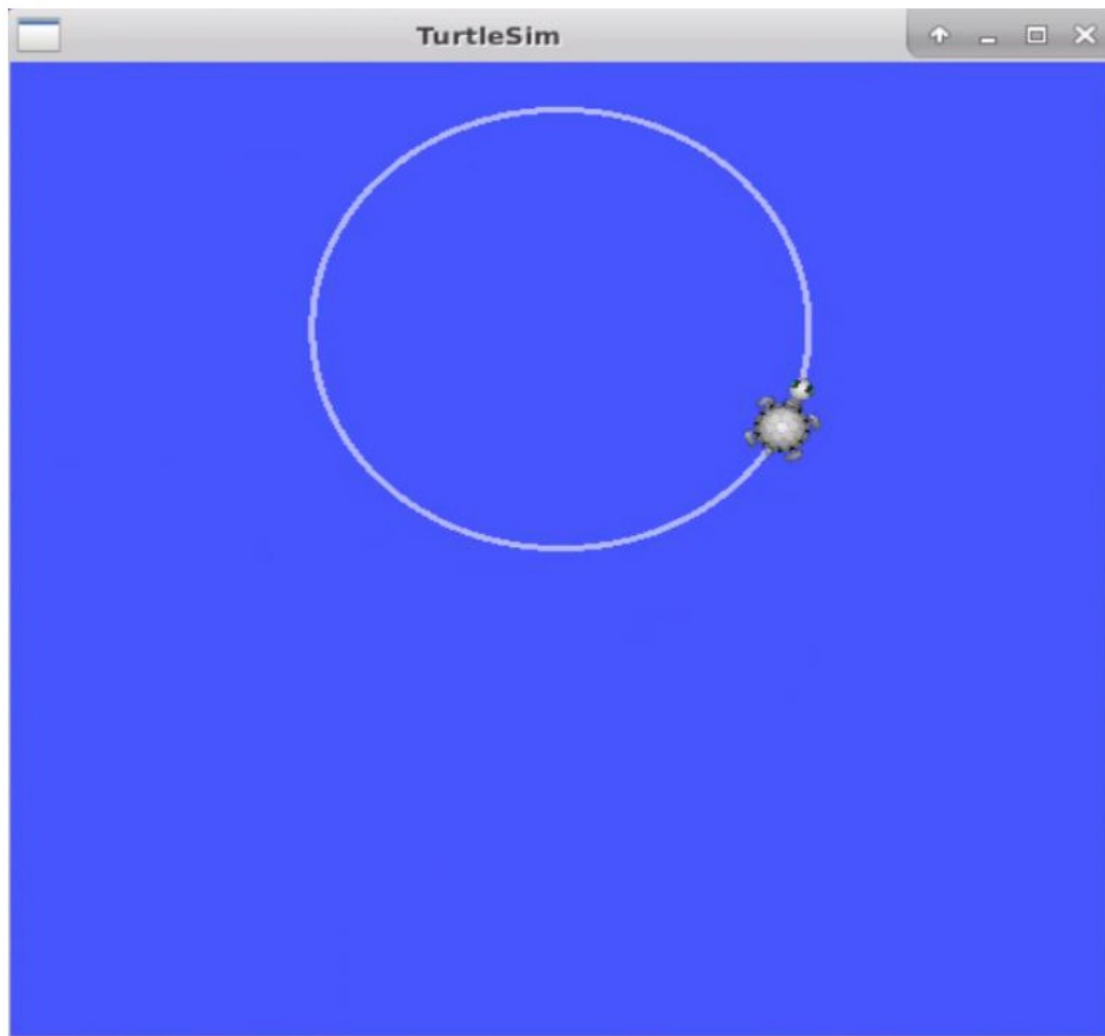


```
ubuntu@ubuntu:~/catkin_ws$ roslaunch turtlesim turtlesim_node
libEGL warning: DRI2: failed to authenticate
[ INFO] [1644663888.439197118]: Starting turtlesim with node name /turtlesim
[ INFO] [1644663888.462207665]: Spawning turtle [turtle1] at x=[5.544445], y=[5.544445], theta=[0.000000]
```



3) 再打开一个新的终端,输入指令“**roslaunch beginner_hiwonder velocity_publisher.py**”
运行velocity_publisher.py的发布者。按下“Ctrl+C”即可停止发布者节点的运行。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roslaunch beginner_hiwonder velocity_publisher.py
[INFO] [1644664960.154518]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664960.255668]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664960.355954]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664960.460197]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664960.556231]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664960.656011]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664960.757576]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664960.863103]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664960.959313]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
[INFO] [1644664961.055680]: Publish turtle velocity command[0.50 m/s, 0.20 rad/s]
```



4) 重新打开一个命令行终端，输入指令

“`roslaunch beginner_hiwonder pose_subscriber.py`” 运行 `pose_subscriber.py` 的订阅者。按下
“**Ctrl+C**” 即可停止订阅者节点的运行。

注意：①启动发布者节点后，订阅者节点才能订阅消息。

②若需要完全接收发布者消息，可以先启动订阅者节点再启动发布者节点。
